

How Circuit Breakers Work

How the circuit breaker works

How It Works in Tyk

- Monitors upstream responses for HTTP status 500+.
- Rate-based: Uses a rolling 10s sample window.
- Failure threshold = failure rate × minimum sample size.

Example:

- Minimum sample size: 100
- Failure rate: 0.5 (50%)
→ Breaker trips if ≥50 failures in last 10s

Post-Trip Behavior

- Breaker opens → returns HTTP 503 to clients.
- Remains open during cooldown period.
- After cooldown, breaker closes and resumes traffic.

Half Open State

Upstream may recover before the cooldown ends.
Full cooldown → unnecessary downtime for clients.

Solution: Half-Open Mode

- Tyk enters half-open state after tripping.
- Client traffic still blocked, but...
- Tyk periodically probes the upstream.

If Probes Succeed:

- Path considered restored.
- Circuit is automatically closed.
- API traffic resumes before cooldown expires.